Ch 13: Our Environment

Question Bank | 2M | 3M | 5M

SECTION A - 2 MARKS

Q1. What is meant by environment? Name any two components of the environment. (PYQ Pattern)

Q2. Define ecosystem. Give two examples of natural ecosystems. (PYQ Pattern)

Q3. What is an artificial ecosystem? Give two examples. (PYQ Pattern)

Q4. What are biotic and abiotic components of an ecosystem? Give one example of each. (PYQ Pattern)

Q5. Define producer. Why are green plants called producers? (PYQ Pattern)

Q6. What are consumers? Name two types of consumers with one example each. (PYQ Pattern)

Q7. What are decomposers? Name any two decomposers. (PYQ Pattern)

Q8. Why are decomposers important in an ecosystem? Give two reasons. (PYQ Pattern)

Q9. What is a food chain? Give one example of a terrestrial food chain. (PYQ Pattern)

Q10. What is a food web? How is it different from a food chain? (PYQ Pattern)

Q11. Define trophic level. What is the first trophic level in a food chain? (PYQ Pattern)

Q12. What are herbivores? Which trophic level do they occupy? Give one example. (PYQ Pattern)

Q13. What are carnivores? Give two examples from a food chain. (PYQ Pattern)

Q14. What are omnivores? Give two examples. (PYQ Pattern)

Q15. State the ten percent law of energy transfer in a food chain. (PYQ Pattern)

Q16. Why is the flow of energy in an ecosystem unidirectional? (PYQ Pattern)

Q17. Why are food chains generally limited to three or four trophic levels? (PYQ Pattern)

Q18. What happens to the amount of energy as we move from one trophic level to the next? (PYQ Pattern)

Q19. What is biological magnification? Name one pollutant that can undergo biomagnification. (PYQ Pattern)

Q20. Why is maximum concentration of harmful chemicals found in top carnivores? (PYQ Pattern)

Q21. Define biodegradable substances. Give two examples. (PYQ Pattern)

Q22. Define non-biodegradable substances. Give two examples. (PYQ Pattern)

Q23. Why do non-biodegradable substances cause more serious environmental problems? (PYQ Pattern)

Q24. What is ozone? Write its molecular formula. (PYQ Pattern)

Q25. Where is the ozone layer present? Why is it important for life on Earth? (PYQ Pattern)

Q26. What are CFCs? How do they affect the ozone layer? (PYQ Pattern)

Q27. What is ozone layer depletion? State one harmful effect of ozone depletion. (PYQ Pattern)

Q28. What is meant by waste management? Why is waste segregation important? (PYQ Pattern)

Q29. Write two methods of managing biodegradable waste. (PYQ Pattern)

Q30. Write two methods of managing non-biodegradable waste. (PYQ Pattern)

Q31. Why should plastic bags be avoided? Give two reasons. (PYQ Pattern)

Q32. What is composting? Which type of waste is suitable for composting? (PYQ Pattern)

Q33. How does excessive use of pesticides affect a food chain? (PYQ Pattern)

Q34. Why are decomposers called natural recyclers? (PYQ Pattern)

Q35. Name the producer and top carnivore in the food chain: grass -> deer -> lion. (PYQ Pattern)

Q36. If 10,000 J energy is available at producer level, how much energy is available to the herbivore level? (PYQ Pattern)

Q37. If 1,000 J energy is available at herbivore level, how much energy reaches the next carnivore level? (PYQ Pattern)

Q38. Why is vegetarian food considered more energy-efficient in terms of food chain energy transfer? (PYQ Pattern)

Q39. What is the role of sunlight in an ecosystem? (PYQ Pattern)

Q40. What is the difference between a natural ecosystem and an artificial ecosystem? Give one example each. (PYQ Pattern)

SECTION B - 3 MARKS

Q1. Explain the main components of an ecosystem with examples of biotic and abiotic components. (PYQ Pattern)

Q2. Describe the role of producers, consumers and decomposers in maintaining ecosystem balance. (PYQ Pattern)

Q3. What are decomposers? Explain how they help in recycling nutrients in nature. (PYQ Pattern)

Q4. Construct a food chain with four trophic levels and identify the producer, primary consumer, secondary consumer and tertiary consumer. (PYQ Pattern)

Q5. Differentiate between food chain and food web on the basis of structure, stability and energy flow. (PYQ Pattern)

Q6. Explain trophic levels in a food chain using a suitable example. (PYQ Pattern)

Q7. State and explain the ten percent law of energy transfer with an example. (PYQ Pattern)

Q8. In a food chain, 20,000 J energy is available at producer level. Calculate energy available at primary consumer, secondary consumer and tertiary consumer levels. (PYQ Pattern)

Q9. Why does the number of organisms decrease at successive trophic levels in a food chain? Explain. (PYQ Pattern)

Q10. Why is the energy flow in an ecosystem considered unidirectional whereas nutrients are recycled? (PYQ Pattern)

Q11. What is biological magnification? Explain it with reference to pesticides entering an aquatic food chain. (PYQ Pattern)

Q12. Why are human beings likely to have higher concentration of non-biodegradable chemicals when they occupy the top position in a food chain? (PYQ Pattern)

Q13. Compare biodegradable and non-biodegradable substances with examples and environmental effects. (PYQ Pattern)

Q14. Explain why non-biodegradable wastes are harmful even if they are produced in small quantities. (PYQ Pattern)

Q15. Describe three harmful effects of using disposable plastic products on the environment. (PYQ Pattern)

Q16. What is ozone layer? Explain how it protects living organisms. (PYQ Pattern)

Q17. Explain how CFCs cause depletion of the ozone layer. Mention one step taken to reduce ozone depletion. (PYQ Pattern)

Q18. What are the harmful effects of ultraviolet radiation reaching the Earth due to ozone layer depletion? (PYQ Pattern)

Q19. Explain the need for waste segregation at source. Classify kitchen waste, paper, plastic and metal cans as biodegradable or non-biodegradable. (PYQ Pattern)

Q20. Explain three methods to manage biodegradable waste at home or school level. (PYQ Pattern)

Q21. Explain three methods to reduce the impact of non-biodegradable waste. (PYQ Pattern)

Q22. What is an aquarium? Why is it considered an artificial ecosystem? Name its biotic and abiotic components. (PYQ Pattern)

Q23. A garden is an ecosystem. Identify its producers, consumers, decomposers and abiotic components. (PYQ Pattern)

Q24. Why should we use paper bags or cloth bags instead of plastic bags? Explain. (PYQ Pattern)

Q25. Explain how overuse of fertilizers and pesticides can disturb an ecosystem. (PYQ Pattern)

Q26. What would happen if all decomposers disappeared from the Earth? Give three effects. (PYQ Pattern)

Q27. What would happen if producers were removed from an ecosystem? Explain the effect on consumers and energy flow. (PYQ Pattern)

Q28. Why is a food web more stable than a single food chain? Explain with an example. (PYQ Pattern)

Q29. Explain the relationship between photosynthesis, producers and energy flow in ecosystems. (PYQ Pattern)

Q30. Give reasons: (i) plants occupy first trophic level (ii) energy decreases at higher trophic levels (iii) top carnivores are fewer in number. (PYQ Pattern)

Q31. Answer the following: (i) What is trophic level? (ii) What is biological magnification? (iii) What are decomposers? (PYQ Pattern)

Q32. Compare natural ecosystem and artificial ecosystem with respect to origin, maintenance and examples. (PYQ Pattern)

Q33. Explain why environment-friendly habits are necessary in daily life. Mention three such habits. (PYQ Pattern)

Q34. Case-based: A pesticide enters a pond ecosystem. Explain how it moves through plankton, small fish, large fish and birds. Which organism will be most affected and why? (PYQ Pattern)

Q35. Case-based: A locality mixes wet waste and plastic waste together. Explain three problems this can create and suggest three corrective measures. (PYQ Pattern)

SECTION C - 5 MARKS

Q1. Describe an ecosystem in detail. Explain its biotic and abiotic components and show how producers, consumers and decomposers interact with each other. (PYQ Pattern)

Q2. Explain the structure and functioning of a food chain. Give one example and identify all trophic levels. Explain how energy flows through the chain. (PYQ Pattern)

Q3. What is food web? Explain why food webs are more stable than food chains. Draw or describe a food web using grass, insects, frog,

snake, eagle, deer and lion. (PYQ Pattern)

Q4. State the ten percent law. A plant has 50,000 J of energy. Calculate energy available to herbivores, primary carnivores and secondary carnivores. Explain why energy decreases at each level. (PYQ Pattern)

Q5. Explain why only a small number of trophic levels are present in a food chain. Include energy loss, respiration, heat loss and availability of biomass in your answer. (PYQ Pattern)

Q6. Write a comprehensive note on producers, consumers and decomposers. Mention their roles, examples and importance in maintaining balance in nature. (PYQ Pattern)

Q7. What are decomposers? Explain their role in nutrient cycling, soil fertility and cleaning the environment. What would happen in their absence? (PYQ Pattern)

Q8. Explain biological magnification in detail. Use an aquatic food chain to show why top carnivores receive the highest concentration of toxic chemicals. (PYQ Pattern)

Q9. Describe how pesticides used in fields can finally affect human beings through food chains. Explain why non-biodegradable chemicals are more dangerous. (PYQ Pattern)

Q10. Compare biodegradable and non-biodegradable wastes in detail. Include definition, examples, decomposition, environmental effect and method of disposal. (PYQ Pattern)

Q11. Explain the problems caused by plastic waste. Mention its effect on soil, water bodies, animals and drainage systems. Suggest measures to reduce plastic pollution. (PYQ Pattern)

Q12. What is ozone layer? Explain the formation and importance of ozone in the upper atmosphere. How does ozone depletion affect living organisms? (PYQ Pattern)

Q13. Explain how CFCs damage the ozone layer. Mention the role of international agreements and public responsibility in protecting the ozone layer. (PYQ Pattern)

Q14. Write a detailed note on waste management. Include segregation, composting, recycling, reuse and safe disposal of non-biodegradable waste. (PYQ Pattern)

Q15. Explain the difference between natural and artificial ecosystems. Use forest, pond, crop field and aquarium as examples. Mention why artificial ecosystems need human maintenance. (PYQ Pattern)

Q16. Describe the flow of energy and cycling of matter in an ecosystem. Why is energy flow unidirectional but materials are recycled? (PYQ Pattern)

Q17. Case Study: In a pond, algae are eaten by small fish, small fish are eaten by large fish, and large fish are eaten by birds. A pesticide enters the pond. Answer: (i) identify trophic levels (ii) explain biomagnification (iii) organism most affected (iv) why pesticide concentration increases (v) one preventive measure. (PYQ Pattern)

Q18. Case Study: A school canteen produces vegetable peels, paper cups, plastic wrappers and metal cans daily. Classify the wastes and prepare a scientific waste-management plan. (PYQ Pattern)

Q19. A food chain is given: grass -> grasshopper -> frog -> snake -> eagle. Answer: (i) identify producer and consumers (ii) state trophic levels (iii) calculate energy reaching eagle if grass has 100,000 J (iv) explain energy loss (v) state one effect if frogs disappear. (PYQ Pattern)

Q20. Explain how human activities disturb ecosystems. Include pesticide use, plastic waste, deforestation, overconsumption and improper waste disposal. Suggest corrective steps. (PYQ Pattern)

Q21. Why is environmental conservation necessary? Explain using examples of food chains, ozone layer protection, waste management and responsible consumer behaviour. (PYQ Pattern)

Q22. Describe an aquarium as an artificial ecosystem. Mention its components, food chain, human maintenance required and what happens if oxygen supply or decomposers are absent. (PYQ Pattern)

Q23. Explain the importance of the 3R principle - reduce, reuse and recycle - in controlling non-biodegradable waste. Give suitable examples for each. (PYQ Pattern)

Q24. Compare food chain, food web and trophic level in a table. Then explain why removal of one species can disturb the entire ecosystem. (PYQ Pattern)

Q25. Write a centum-level answer on 'Our Environment' covering ecosystem, trophic levels, energy transfer, biomagnification, ozone layer and waste management. (PYQ Pattern)

Q26. Answer the following in detail: (i) Why are producers essential? (ii) Why are decomposers essential? (iii) Why are non-biodegradable chemicals dangerous? (iv) Why should waste be segregated? (v) Why is ozone layer protection important? (PYQ Pattern)